

Artificial Intelligence

CS471-GR2 Workshop Review

Instructions: Break into teams as determined by your instructor. Each team will be assigned a question or two to discuss and answer as a group. After 25 minutes each group will address the class explaining their answers. Focus on explaining the underlying concepts, comparing algorithmic approaches, and applying these theories to practical scenarios.

Lesson 11 — Markov Decision Processes I

- 11.1 Define the foundational components of a Markov Decision Process. How do these elements sequentially interact to formalize an agent's decision-making environment?
- 11.2 What is the Markov property, and why is this assumption critical for making sequential decision-making problems computationally tractable?
- 11.3 Formulate a complex operational scenario (e.g., autonomous drone navigation, inventory management) strictly as an MDP. Clearly designate what constitutes the state space, the action space, the transition probabilities, and the reward structure.

Lesson 12 — MDP II

- 12.1 What is the conceptual distinction between a policy and a value function? How is a value function utilized to formally evaluate the effectiveness of a specific policy?
- 12.2 How is expected cumulative reward calculated over time? Explain the role of the discount factor and describe how an agent's behavior shifts when the discount factor approaches 0 versus when it approaches 1.
- 12.3 What specific conditions must be met for a policy to be considered optimal? How does the optimal value function mathematically relate to this optimal policy?

Lesson 13 — Reinforcement Learning I (Value Iteration)

- 13.1 Describe the iterative mechanics of the Value Iteration algorithm. How does it iteratively refine estimates of state values from initialization to convergence?
- 13.2 Write out the Bellman update equation and describe each of its variables. How does this mathematical update link a single state's value to the expected values of its subsequent states?

- 13.3** What does it mean for the value function to “converge” during Value Iteration? Why is convergence to the true optimal value mathematically guaranteed as iterations approach infinity?

Lesson 14 — Reinforcement Learning II (Policy Iteration)

- 14.1** Break down the Policy Iteration algorithm into its core phases: policy evaluation, policy extraction, and iteration. What specific mathematical computations occur during each of these phases?
- 14.2** When applying policy iteration to a simple grid-world decision problem, how do you extract a new, updated policy from the utilities calculated during the evaluation step?
- 14.3** Compare and contrast Policy Iteration with Value Iteration. Under what specific environmental constraints or state-space sizes might one algorithm be practically preferred over the other?

Lesson 15 — Reinforcement Learning III (Q-Learning)

- 15.1** What is the fundamental difference between model-based and model-free reinforcement learning paradigms? Describe a real-world scenario where model-free learning is strictly necessary.
- 15.2** Explain the concept of Q-learning and define what a Q-value represents. How does this algorithm utilize temporal difference methods to update values without requiring a known transition model?
- 15.3** Interpret how an agent learns from raw experience in Q-learning. Why is balancing exploration and exploitation critical to accurately learning the optimal Q-values over time?